

Your grade: 100%

Next item →

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1.

Let $A = \{a, b\}$ and let $\mathcal{R}$ be a binary relation on a set A defined by $\mathcal{R} = \{(a, a), (a, b), (b, b)\}$. Which one the following statements is correct?

1 / 1 point
- ☐

$\mathcal{R}$ is not an equivalence relation as it is not transitive

☒

$\mathcal{R}$ is not an equivalence relation as it is not symmetric

☐

$\mathcal{R}$ is not an equivalence relation as it is not reflexive

☐

$\mathcal{R}$ is an equivalence relation

👍 Nice work

Correct. $\mathcal{R}$ is not symmetric as $(a, b) \in \mathcal{R}$ but $(b, a) \notin \mathcal{R}$.

2.

Let $A = \{a, b, c\}$ and let $\mathcal{R}$ be a binary relation on a set A defined by $\mathcal{R} = \{(a, a), (b, b), (b, c), (c, c)\}$. Which one the following statements is correct?

1 / 1 point
- ☐

$\mathcal{R}$ is not an equivalence relation as it is not transitive

☒

$\mathcal{R}$ is not an equivalence relation as it is not symmeric

☐

$\mathcal{R}$ is not an equivalence relation as it is not reflexive

☐

$\mathcal{R}$ is an equivalence relation

👍 Nice work

Correct. $\mathcal{R}$ is not symmetric as $(b, c) \in \mathcal{R}$ but $(c, b) \notin \mathcal{R}$.

3.

Let $\mathcal{R}$, a binary relation on $\mathbb{Z}$ (a set of all integers), be defined as $a\mathcal{R}b$ if and only if $a \leq b, \forall a, b \in \mathbb{Z}$. Which one of the following statements is true about the relation $\mathcal{R}$?

1 / 1 point
- ☐

$\mathcal{R}$ is not an equivalence relation as it is not anti-symmetric

☐

$\mathcal{R}$ is an equivalence relation

☒

$\mathcal{R}$ is not an equivalence relation as it is not symmetric

☐

$\mathcal{R}$ is not an equivalence relation as it is not transitive

👍 Nice work

Correct. $\mathcal{R}$ is not symmetric as $2 \leq 3$ but $3 \not\leq 2$.

4.

Let $\mathcal{R}$, a binary relation on $\mathbb{Z}$ (a set of all integers), be defined as $a\mathcal{R}b$ if and only if $(a + b) \bmod 2 = 0, \forall a, b \in \mathbb{Z}$. Which one of the following statements is true about the relation $\mathcal{R}$?

1 / 1 point
- ☐

$\mathcal{R}$ is not an equivalence relation as it is not transitive

☐

$\mathcal{R}$ is not an equivalence relation as it is not symmetric

☒

$\mathcal{R}$ is an equivalence relation

☐

$\mathcal{R}$ is not an equivalence relation as it is not reflexive

👍 Nice work

Correct. $\mathcal{R}$ is reflexive, symmetric, and transitive.

5.

Let $\mathcal{R}$, an equivalence binary relation on $\mathbb{Z}$ (a set of all integers), be defined as $a\mathcal{R}b$ if and only if $(a \bmod 2 = b \bmod 2, \forall a, b \in \mathbb{Z})$. How many equivalence classes are in $\mathcal{R}$?

1 / 1 point
- ☐

1

☐

3

☐

4

☒

2

👍 Nice work

Correct. There are 2 equivalence classes. The first equivalence class contains all odd numbers as they are related to each other, and the second class contains all even numbers as they are related to each other.

6.

Let $A = \{a, b, c, d\}$ and let $\mathcal{R}$ be an equivalence relation on a set A defined by $\mathcal{R} = \{(a, a), (b, b), (a, b), (b, a), (c, c), (d, d)\}$. How many equivalence classes are in $\mathcal{R}$?

1 / 1 point
- ☐

0

☒

3

☐

1

☐

2

👍 Nice work

Correct. There are three equivalence classes; $\{a, b\}$, $\{c\}$ and $\{d\}$.

7.

Let $A = \{1, 2, 3, 4\}$ and let $\mathcal{R}$ be an equivalence relation on a set A defined by $\mathcal{R} = \{(a, b) \mid a, b \in A \text{ and } (a + b) \bmod 2 = 0\}$. Which one of the following is true about $\mathcal{R}$?

1 / 1 point
- ☐

$\mathcal{R}$ has three equivalence classes, $\{1\}$, $\{2\}$ and $\{3\}$

☐

$\mathcal{R}$ has one equivalence class, $\{1, 2, 3\}$

☐

$\mathcal{R}$ has no equivalence class

☒

$\mathcal{R}$ has two equivalence classes, $\{1, 3\}$ and $\{2, 4\}$

👍 Nice work

Correct. Even numbers are related to each other, and odd numbers are related to each other.